

Causal Inference



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THE REMIX

Advanced Topics in Applied Causal Inference

A test of the Remix style

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LESSON

1. DiD Origins

The four averages and three subtractions

Ashenfelter's move at Princeton

- DiD is not a regression. It's an **idea**.
- Orley Ashenfelter at the Industrial Relations Section: applied research questions, communicate to bureaucrats, get the answer right with high-quality data.
- His move: don't emphasize the regression — emphasize the simplicity of **four averages and three subtractions**.
- The 2×2 table: treated/control by pre/post; four cells; four averages.

The DiD formula

$$\widehat{\text{DiD}} = \underbrace{(Y_{11} - Y_{10})}_{\text{treated change}} - \underbrace{(Y_{01} - Y_{00})}_{\text{control change}}$$

Block styles

Three flavors for three uses

Standard block

Body text uses slate. Title is charcoal on a light background. For neutral commentary or definitions.

Alert block

For **key insights** or warnings. The accent color tints the background subtly. Use sparingly — restraint matters.

Example block

For worked examples or empirical illustrations. Ocean-blue tint distinguishes it from the alert.

Voice of the field

If you can convince a bureaucrat in five minutes, you have done something. Anything else is academic.

— Orley Ashenfelter (paraphrased, Scott's creative license)

The point is communication. The 2×2 doesn't need a regression to tell its story.

LESSON

2. The Three Regressions

Equivalence on castle.dta

2005–2006 subset, one DiD, three ways

Three specifications:

1. OLS with interactions

```
reg l_homicide post##treat
```

2. TWFE

```
xtreg ..., fe vce(cluster sid)
```

3. Long difference

```
reshape wide; reg diff treat
```

All three return the same number to machine epsilon.

This equivalence is what **made it possible** to extend DiD to covariates, staggered, and continuous treatment using OLS infrastructure.

But the equivalence breaks once you add complexity. Day 2.